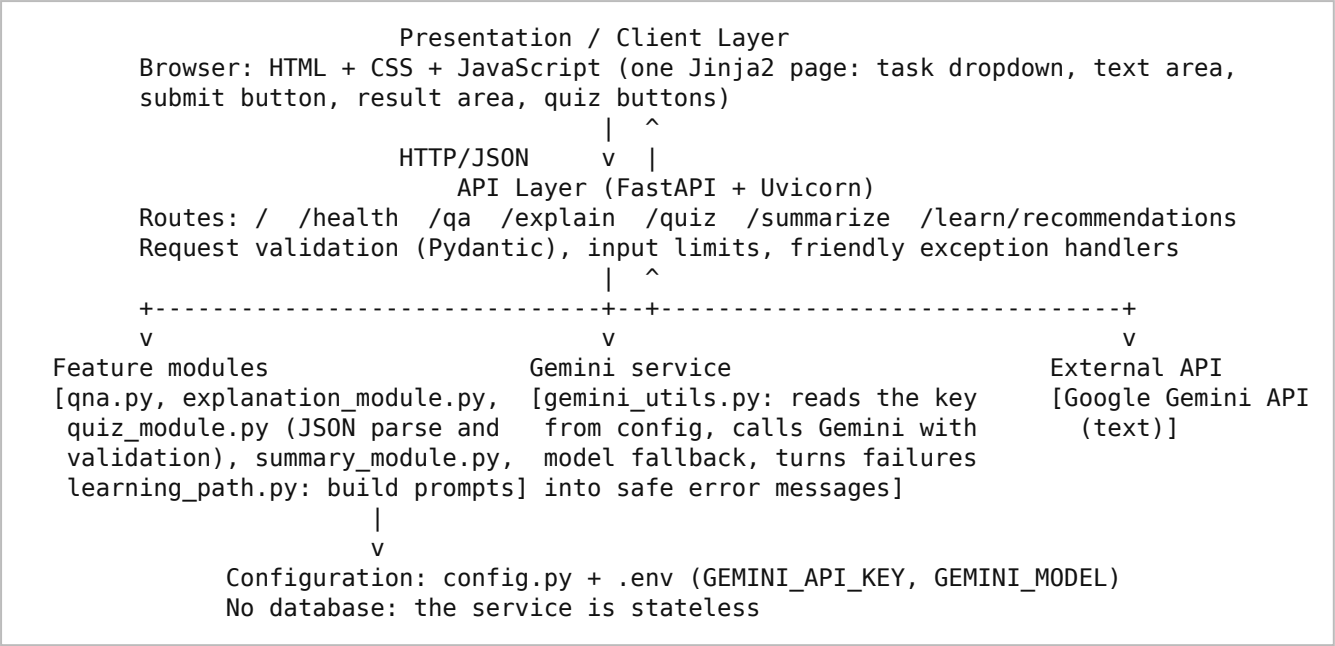


Solution Architecture

Date	08 October 2026
Team ID	SWTID-2026-2386
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	One responsive page: choose a task, type input, read the result, answer quiz questions with instant feedback	HTML5, CSS3, JavaScript, Jinja2
API Layer (main.py)	Routing, validation, input limits and friendly error handling; one endpoint per feature, plus /health	FastAPI, Uvicorn, Pydantic
Models (models.py)	Request schema shared by every endpoint	Pydantic
Feature modules	One file per feature builds the prompt: qna.py, explanation_module.py, summary_module.py, quiz_module.py (also cleans and validates the JSON), learning_path.py	Python
Gemini service (gemini_utils.py)	Calls Gemini, tries fallback models if one is retired or busy, and converts failures into messages that are safe to show	google-genai
Configuration (config.py)	Reads the API key, model name, timeouts and limits from environment variables	python-dotenv
External APIs	Google Gemini for all generated content	Google Gemini API